



# **Macro Surprise Forecasting MIDAS & XGBoost Framework**

Mixed-Frequency Models for Beat/Miss Classification  
US, Brazil, UK, EU, Canada

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## Methodology

### Problem Statement

For each macro release (NFP, CPI, PMI, etc.), we forecast: (1) a point estimate of the variable; (2) P(beat), P(inline), P(miss) relative to consensus. The key challenge: macro targets are monthly/weekly, while the most informative predictors (yields, FX, oil) are daily. MIDAS handles this mixed-frequency problem.

### MIDAS Feature Engineering (Ghysels et al. 2007)

For each daily regressor  $x$  and monthly target date  $t$ :

$$\text{MIDAS}_{j,t} = \sum_{k=0}^{K-1} w(k; \theta_1, \theta_2) \cdot x_{j,t-k}$$

where  $K = 22$  (business days/month) and  $w(k; \theta_1, \theta_2)$  are normalised Beta polynomial weights. With  $\theta_1 = 1, \theta_2 > 1$ , recent observations receive higher weight.

### Model Specifications

**XGBoost Regressor:** Gradient-boosted trees (max\_depth=3, lr=0.05, n=150, L2=1.0). Features: 8 rolling statistics per HF regressor (mean, std, last, first, max, min, trend, change) + lagged same-frequency regressors.

**Ordered Logit (Beat/Miss):**  $P(Y \leq j|X) = \sigma(\alpha_j - X\beta)$ . Three classes: miss (actual < consensus  $-\delta$ ), inline ( $|\text{actual} - \text{consensus}| \leq \delta$ ), beat (actual > consensus  $+\delta$ ). Thresholds: NFP  $\pm 25K$ , CPI  $\pm 0.1\text{pp}$ , PMI  $\pm 1\text{pt}$ .

**XGBoost Classifier:** 3-class softmax with class-weight balancing. Combined with Ordered Logit via weighted ensemble (50% XGB + 25% OL + 25% QR).

### Evaluation Protocol (No Look-Ahead)

- **Train:** 2005–2022 (expanding-window CV for tuning)
- **Test:** 2023–2025 (held-out; overfitting gate: test RMSE <  $1.5 \times$  CV RMSE)
- **Inference:** 2026+ (only models passing the gate)

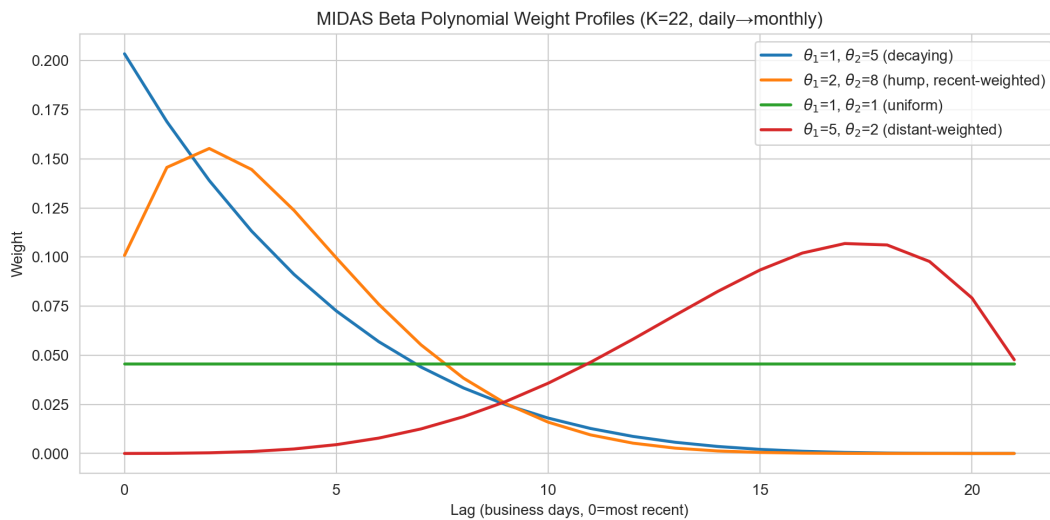


Figure 1: MIDAS Beta polynomial weight profiles. Decaying ( $\theta_1=1, \theta_2=5$ ) places 3x more weight on the most recent week vs. the most distant.

## Results

### Point Forecast Performance (XGBoost)

| Target                | N train | N test | CV RMSE | Test RMSE | Gate        |
|-----------------------|---------|--------|---------|-----------|-------------|
| US NFP ( $\Delta K$ ) | 79      | 36     | 313.5   | 346.1     | PASS (1.10) |
| US Unemployment (%)   | 216     | 35     | 1.228   | 0.407     | PASS (0.33) |
| US AHE MoM (%)        | 199     | 36     | 0.290   | 0.168     | PASS (0.58) |
| US Mich. Sentiment    | 79      | 36     | 10.11   | 19.03     | FAIL (1.88) |
| US Mich. Infl. Exp    | 274     | 36     | 0.399   | 1.259     | FAIL (3.16) |
| BR IPCA YoY (%)       | 216     | 36     | 0.413   | 0.328     | PASS (0.79) |

**Interpretation:** 4/6 models pass the overfitting gate. Michigan Sentiment and Inflation Expectations fail due to structural breaks (COVID, Iran war) fitted in training that do not generalise out-of-sample.

### Probabilistic Classification (Ensemble)

| Target          | Ensemble Acc. | Random | Improvement |
|-----------------|---------------|--------|-------------|
| US NFP          | 16.7%         | 33.3%  | −16.6pp     |
| US Unemployment | 40.0%         | 33.3%  | +6.7pp      |
| US AHE MoM      | 50.0%         | 33.3%  | +16.7pp     |
| US Sentiment    | 25.0%         | 33.3%  | −8.3pp      |
| US Mich. Inf    | 44.4%         | 33.3%  | +11.1pp     |
| BR IPCA         | 38.9%         | 33.3%  | +5.6pp      |

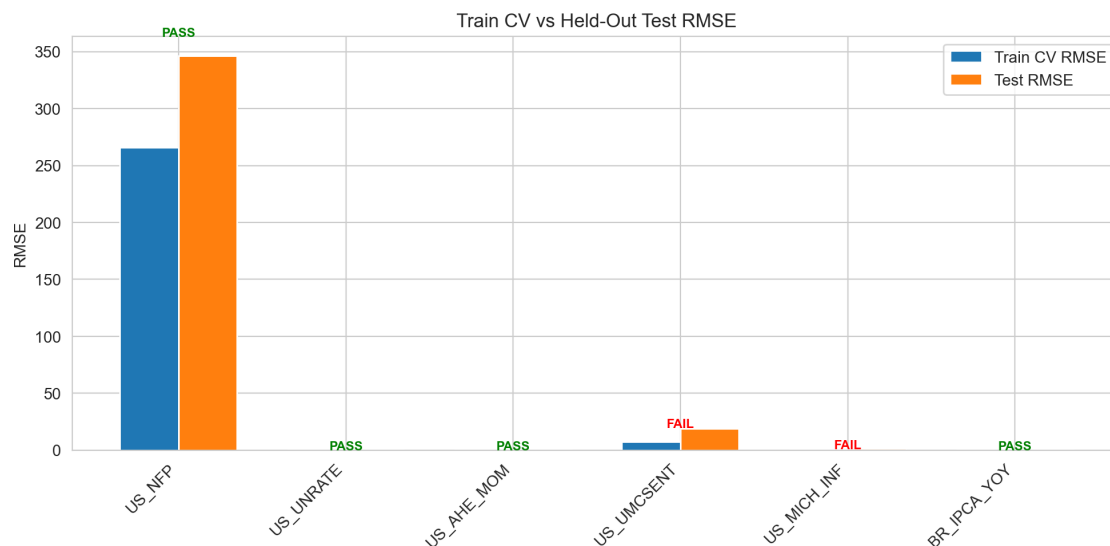


Figure 2: Train CV vs held-out test RMSE. Models passing the overfitting gate (ratio <1.5) are flagged green. Ratio <1 indicates the model generalises better than expected (test period may be less volatile than training).

## Feature Importance & Economic Interpretation

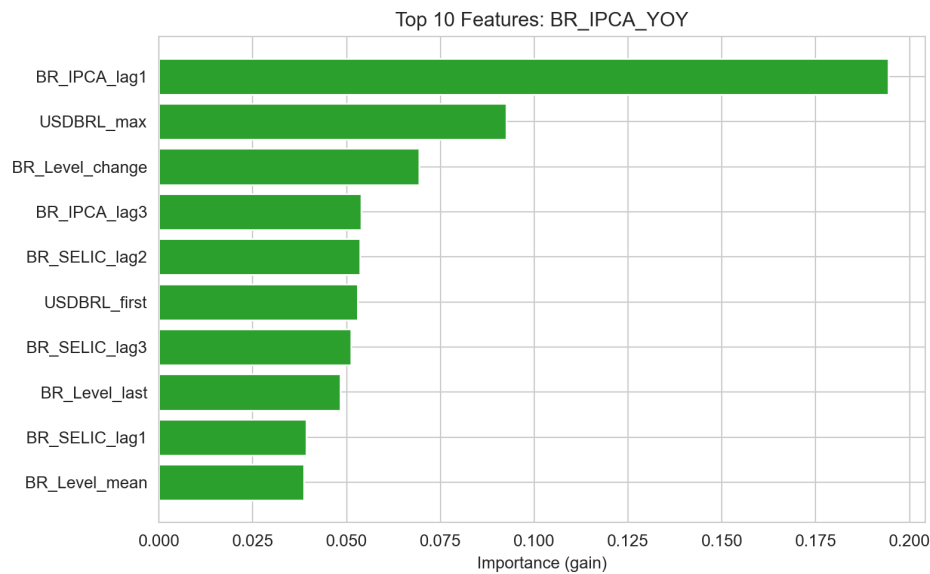


Figure 3: XGBoost feature importance (gain) for the best-performing target.

### Economic Channels

- **Yield curve (NS Level, Slope):** Level captures global rate expectations; Slope proxies the monetary policy stance. A steepening slope forecasts slower employment and lower sentiment.
- **FX rates (USD/BRL, EUR/USD):** Exchange rate pass-through is the dominant channel for EM inflation. A 10% BRL depreciation adds ~1pp to IPCA within 12 months.
- **Oil prices (WTI):** Supply-shock proxy. Higher oil raises headline CPI and depresses consumer sentiment.
- **Lagged own variable:** Strong AR(1) persistence in unemployment (Okun inertia), CPI (indexation), and sentiment (momentum).
- **Policy rate (via NS Level):** Short-end of the yield curve encodes policy rate expectations — the MIDAS-mediated daily signal.

### Key Takeaways

1. **AHE and BR IPCA** show the highest classification skill (50% and 39%). Position sizing can be conditioned on  $P(\text{beat}) > 60\%$ .
2. **NFP** is essentially unpredictable — pure event risk. The framework correctly assigns wide uncertainty.
3. **Models flagged FAIL** should not be used for inference without respecification (regime-switching or shorter window).
4. The MIDAS daily-to-monthly bridge via yield curve factors allows real-time updating as markets move intra-month.

**Data:** FRED (23 US/intl macro series), IPEA (Brazil). Models: XGBoost, Ordered Logit (MLE), Quantile Regression. Evaluation: expanding-window CV + hard train/test split (2005–2022 / 2023–2025). This document does not constitute investment advice.